

Research Article

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Improving Reading Comprehension of Expository Texts through Text-Structure-Aligned Graphic Organizers for Higher Education Students

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Abstract

Background/purpose. Academic reading presents significant challenges for English for Specific Purposes (ESP) learners, especially when processing structurally dense expository texts in STEM disciplines. This study aims to assess the impact of text-structure-aligned goals on students' reading comprehension and written summarization, while exploring their perceptions of using them as a cognitive and strategic learning tool.

Materials/methods. This research employed an explanatory sequential mixed-methods design involving 200 Indonesian university students in Information Systems and Informatics programs. Students completed expository reading tasks using either T-Charts (cause-and-effect texts) or Venn Diagrams (compare-and-contrast texts), followed by summary writing, which was evaluated using a four-dimensional rubric: accuracy, paraphrasing, focus, and language structure. Open-ended questionnaires captured qualitative data on student perceptions.

Results. Quantitative findings revealed significantly higher comprehension scores with T-Charts ($M = 3.67$) compared to Venn Diagrams ($M = 3.24$), with a medium effect size (Cohen's $d = 0.56$). Rubric-based analysis showed strengths in focus and accuracy, but persistent weaknesses in paraphrasing. Qualitative data indicated that students found GOs helpful for visualizing textual relationships, improving focus, and enhancing information retention.

Conclusion. The study confirms the instructional value of aligning graphic organizers with text structure in ESP contexts. It highlights the need for explicit linguistic support, particularly for paraphrasing, and contributes practical and theoretical insights to academic reading pedagogy grounded in Dual Coding and Schema Theory.

1. Introduction

Academic reading skills are essential for success in the disciplines of Science, Technology, Engineering, and Mathematics (STEM) and English for Specific Purposes (ESP). They allow learners to engage with complex expository texts and comprehend them critically. EFL/ESP students must effectively process specialized information, which necessitates the ability to navigate diverse text structures, particularly those found in academic literature (Bensaad & Ghania, 2020). Regrettably, numerous students face substantial obstacles due to their inadequate comprehension and application of critical concepts essential for professional practice and higher education (Hezam et al., 2022). Additionally, EFL learners frequently encounter difficulties with their fundamental reading abilities due to inadequate preparation in academic reading strategies. This not only restricts their comprehension of course material but also affects their overall academic performance (Rahim et al., 2023).

Additionally, the instructional methods currently implemented in numerous developing countries often fail to incorporate a structured approach to visual strategies, which can help students comprehend complex text structures. Research has demonstrated that visual aids can provide learners with substantial assistance in comprehending the organizing principles of expository texts, thereby enhancing their reading comprehension (Hall-Mills & Marante, 2025; Oteda et al., 2021). Nevertheless, the absence of such strategies in ESP curricula can result in a lacuna that can exacerbate student underachievement. There is an urgent need for a pedagogical framework that integrates systematic visual strategies into academic reading instruction (Al-Malki et al., 2022). By implementing such measures, learners in EFL/ESP not only resolve the structural comprehension issues they encounter, but they also engage more deeply with academic material, thereby improving their chances of success in STEM fields.

Graphic organizers (GOs) are increasingly recognized as essential instructional tools for promoting reading comprehension, particularly in expository texts that present structural challenges for EFL learners. GOs help learners organize complex knowledge and understand abstract connections by visualizing linkages between concepts, such as comparisons using Venn diagrams or causal interactions using T-Charts. Dual Coding Theory posits that integrating verbal and visual information enhances cognitive processing and memory retention (Luo, 2023), while Schema Theory suggests that comprehension improves when new input aligns with learners' prior knowledge structures (Alfares, 2019). Koerich Rondon & Braga Tomitch (2020) stressed the role of GOs as pre-reading tools that activate formal schemata and scaffold comprehension of complex argumentative texts. GOs clarify and sequence academic content dynamically, especially in multilingual or low-proficiency environments (Alfares, 2019), improving student engagement and retention. Kurniaman et al. (2020) found that GOs enhance academic performance and foster higher-order thinking skills necessary for technical and academic genres.

Although the majority of current research on instructional scaffolds, such as GOs, has concentrated on general or K–12 populations, associated instructional strategies like gamification have also become prominent in primary and secondary education owing to their beneficial effects on engagement, motivation, and learning outcomes (Luo, 2023; Vrcelj et al., 2023; Zhang & Hasim, 2023). These methodologies typically incorporate gamified components, such as points, leaderboards, and instant feedback, to enhance student engagement. Nonetheless, despite their established advantages, research on gamification and visual organizers remains constrained within the realm of English for Specific Purposes (ESP), particularly in technical academic publications. Despite advancements in engineering and psychology (Lyu, 2024; Mostafavi et al., 2021), further investigation into discipline-specific and genre-based uses of visual methods in ESP training remains necessary (Hansen, 2022). Recent studies indicate that GOs enhance critical thinking in EFL reading (Min et al., 2023), suggesting that their integration into ESP contexts, particularly in STEM subjects,

necessitates further empirical inquiry to maximize strategic reading in highly structured, domain-specific texts.

Despite the growing body of research on the use of graphic organizers in reading instruction, numerous critical gaps remain. The congruence between specific expository text structures, such as cause and effect or compare and contrast, and the corresponding graphic organizer formats designed to scaffold them effectively has been rarely examined in existing studies (Alfares, 2019; Qi & Jiang, 2021). In particular, there has been a lack of focus on the evaluation of students' linguistic performance using rubric-based assessments, particularly in paraphrasing, which necessitates targeted and explicit instruction (Stevens et al., 2020; Yahia & Egbert, 2023). Prior research has demonstrated that paraphrasing is one of the most difficult academic skills for EFL learners and cannot be fully addressed by visual tools alone without explicit linguistic support (Flores & Lopez, 2019; Parameswari et al., 2024).

Qualitative research on students' graphic organizer use in technical fields based on English as a Foreign Language are scarce. Specialized language and sophisticated conceptual structures make comprehension harder in these circumstances (Min et al., 2023). This study integrates Dual Coding Theory, which posits that information is better understood and retained when presented in both verbal and visual forms, and Schema Theory, which emphasizes that comprehension improves when new information is connected to learners' existing knowledge frameworks (Maxim & Brown, 2023). These theoretical perspectives directly inform the research questions: the study examines how structured instructional tools support comprehension (Dual Coding) and activate prior knowledge schemas (Schema Theory), operationalized through measurable outcomes in summary writing, including accuracy, paraphrasing, focus, and language structure. This study will be employed an explanatory sequential mixed-methods design, collecting quantitative data on students' reading comprehension through structured tasks and analyzing the results using descriptive statistics, paired-sample t-tests, and effect size calculations (Cohen's *d*). Qualitative data were gathered via open-ended questionnaires and analyzed through thematic coding to explore students' perceptions and learning experiences.

To address these theoretical and practical concerns, this study is guided by three key research questions: (1) How do graphic organizers affect students' reading comprehension performance in summarizing expository texts? (2) What are the patterns of student performance across different dimensions of summary writing? and (3) What are students' perceptions and learning experiences when using graphic organizers? These questions frame the investigation into how graphic organizers function not only as instructional tools to enhance comprehension, but also as cognitive and strategic supports in academic reading for students in technical education contexts.

2. Literature Review

2.1. Expository Texts and Reading Challenges in EFL

Expository texts, especially for EFL teachers, help spread information and improve understanding for academic success. Expository texts aim to inform rather than entertain. They present facts through explicit definitions, descriptions, and explanations, often arranged using cause-and-effect and compare-and-contrast methods (Balepur et al., 2023). These texts contain specialized knowledge and stimulate higher-order thinking, requiring EFL learners to synthesize information and make inferences, notably in technology and informatics. Expository works vary in structure depending on their purpose. The compare-contrast structure helps readers analyze concepts, improving their analytical skills (Liu, 2021). However, the cause-effect framework helps students understand how events and concepts relate, improving critical thinking (Gazali & Baa, 2023). Structured presentations organize concepts and present information in a methodical manner, thereby improving student engagement and comprehension.

Despite the inherent advantages of expository texts, EFL learners frequently encounter cognitive and linguistic difficulties while reading them. Limited vocabulary, insufficient previous knowledge, and conceptual complexity hinder comprehension (Ahmed, 2021; Yang, 2022). The existence of novel linguistic structures may hinder fluency, hence exacerbating reading sessions (Jusoh & Kassim, 2023; Yang, 2022). Thus, these constraints may hinder expository reading comprehension, prompting teachers to employ effective reading strategies to enhance comprehension and memory (Gazali & Baa, 2023). Students in technology and informatics are affected by issues with reading expository texts. Publications with specialist language and precise information, which are crucial for practical applications, require advanced comprehension skills. Strong readers can better understand and retain expository literature (Ahmed, 2021). Interventions that emphasize metacognitive strategies may reduce the cognitive load of reading, helping students engage more deeply with the content and improve their comprehension. These students require customized support from educators, utilizing digital resources that align with their comprehension levels and structured instructional practices that enhance text engagement and understanding (Wimmer et al., 2023).

In conclusion, acknowledging the distinct attributes and frameworks of expository texts enables instructors to more effectively support EFL learners, particularly those engaged in technical disciplines. By cultivating effective reading techniques through focused instruction and technological integration, educators can significantly enhance comprehension and engagement. This method not only helps pupils overcome reading difficulties but also enables them to achieve academic excellence in their chosen disciplines.

2.2. Graphic Organizers as Scaffolding Tools

Graphic organizers (GOs) are visual tools designed to aid in the arrangement and synthesis of information, providing learners with a systematic approach to comprehend complex concepts and relationships. Fundamental types of graphic organizers encompass Venn diagrams, T-charts, and concept maps, each fulfilling distinct instructional objectives by visually illustrating essential concepts, similarities, and hierarchies (Qi & Jiang, 2021). The theoretical foundations of graphic organizers are linked to Dual Coding Theory and Schema Theory. Multiple Coding Theory asserts that information retention is improved when it is encoded both audibly and graphically, enabling pupils to build multiple representations of knowledge that facilitate learning (Koerich Rondon & Braga Tomitch, 2020). Schema Theory emphasizes the impact of preexisting knowledge on new learning, with graphic organizers serving to activate existing schemas and therefore enhance the assimilation of new information. These findings suggest that graphic organizers (GOs) improve student understanding, critical thinking, and retention, especially in EFL. Visual organizers improve reading comprehension, especially among EFL students, according to multiple studies. Research shows that GOs improve EFL learners' reading comprehension (Barashid, 2020). Graphic organizers help students understand relationships and distinctions between concepts and text structures, which is crucial for understanding complex parts (Koerich Rondon & Braga Tomitch, 2020). The research also shows that GOs encourage cooperative learning by creating an interactive environment where students assess and analyze textual information, improving their grasp of the subject (Eldin et al., 2020). Visual organizers help students overcome linguistic and cognitive challenges in reading comprehension, especially in EFL training.

2.3. Summarization as a Measure of Reading Comprehension

Reading comprehension is measured by students' ability to summarize key information from books and express it in their own words. Summarizing material for reading comprehension requires students to identify core ideas, pertinent details, and unnecessary information. Several ways assist students in summarizing readings. Explicit instructions and structure help students create stronger summaries. Summaries with graphic organizers (GOs) assist pupils in organizing and visualizing

information (Balepur et al., 2023; Khezlrou, 2021). In ESP reading programs, where specialized knowledge is vital for academic success, GOs with summarizing techniques boost student attention and understanding (Hezam et al., 2022). Summary strategies boost reading comprehension and engagement, according to study. The GIST method has increased EFL students' summary writing and reading comprehension (Ayyub & Tyas, 2023; Liu, 2021). Student summarization rubrics offer clear feedback, facilitating students' growth (Hall-Mills & Marante, 2025). These summarizing assignments are popular with students who demonstrate improved comprehension and expression of complicated materials (Pratiwi & Mayrita, 2023). EFL teachers can improve students' comprehension and performance with systematic summaries and GOs.

3. Methodology

3.1. Research Design

This study employed an explanatory sequential mixed-methods design (Åkerblad et al., 2021), as outlined by Creswell and Plano Clark (2018). In this approach, quantitative data were collected and analyzed in the first phase, followed by qualitative data to explain and elaborate on the initial findings. The primary aim of the study was to investigate the effectiveness of using graphic organizers (GOs), specifically Venn Diagrams and T-Charts, in enhancing students' comprehension of expository texts related to information technology. The quantitative component measured students' reading comprehension through structured summarization tasks, while the qualitative component explored students' perceptions and experiences with the use of GOs in the reading process. This design allowed for a comprehensive understanding of both the measurable outcomes and the subjective experiences of learners (McKim, 2017).

3.2. Participants

The participants consisted of approximately 200 undergraduate students enrolled in the Information Systems and Informatics Engineering programs at Universitas Duta Bangsa Surakarta, Indonesia. The purposive sampling technique was employed to select students in their second to third semester of study (Bostley & Peters, 2023). The purpose of this sampling method was to guarantee that the participants had an adequate amount of exposure to both technology-related content and English-language instructional materials, thereby preparing them to engage with expository texts in an English for Specific Purposes (ESP) context. The research was conducted in a formal academic environment, as part of English reading and writing instruction that incorporated content from the students' disciplinary backgrounds.

3.3. Research Instruments

This investigation implemented three instruments: (1) graphic organizer assignments embedded in expository texts, (2) a summarization rubric, and (3) a student response questionnaire. The expository texts, organized as compare-contrast and cause-and-effect, were paired with suitable graphic organizers to assist students in identifying and mapping key ideas during the reading process. The rubric focused on four key areas: accuracy, paraphrasing, focus, and language structure (Hall-Mills & Marante, 2025; Min et al., 2023; Wong & Samudra, 2021). To evaluate the results of reading comprehension, each aspect was assessed on a scale of 1 to 5. A questionnaire that integrated open-ended questions and Likert-scale items was implemented to investigate student perceptions. It obtained students' perspectives regarding the lucidity, utility, and obstacles associated with employing graphic organizers in reading assignments.

Table 1. Instrument Specification and Blueprint

Instrument	Dimension / Aspect	Indicator	Scale / Format
Expository Text + GO Task	Text structure recognition	Student identifies structure (compare–contrast or cause–effect)	Graphic Organizer
	Content mapping	Student maps ideas into correct categories	Open response (visual)
Summarization Rubric	Accuracy	Includes key points from the text correctly	1–5 Likert scale
	Paraphrasing	Uses own words, avoids direct copying	1–5 Likert scale
	Focus and relevance	Summary stays on topic and includes main ideas	1–5 Likert scale
	Language structure	Coherent, grammatically appropriate expression	1–5 Likert scale
	Usefulness of GO	“The graphic organizer helped me understand the text better.”	Likert-scale (1–5)
Student Response Questionnaire	Clarity of use	“The instructions for using the GO were clear and easy to follow.”	Likert-scale (1–5)
	Reflection on experience	Open-ended: “What did you find helpful or difficult in using GO?”	Open response

3.4. Procedures

The study was conducted over approximately six sessions, each lasting 90 minutes. The overall procedure can be summarized as follows:

- Orientation session: Students were introduced to the purpose of the study and the general structure of expository texts.
- Instructional sessions: Two sessions focused on different text structures. In each session, students read assigned texts, engaged with structured tasks to organize information, and wrote summaries based on their notes.
- Data collection sessions: The remaining sessions were used to collect students’ summaries and administer the perception questionnaire.

This structured approach allowed students to engage with the texts progressively, while enabling the researchers to collect both quantitative and qualitative data for subsequent analysis.

3.5. Data Analysis

Quantitative data from students' summary scores were analyzed using descriptive statistics (mean, median, and mode) to identify patterns in reading comprehension. The efficacy of the graphic organizer intervention was assessed with paired-sample t-tests, and Cohen's d was calculated to determine the effect size and practical significance of observed improvements. Prior to conducting these parametric analyses, the normality of all quantitative variables was evaluated using the

Shapiro-Wilk test, confirming that all data were normally distributed ($p > 0.05$) and that the chosen statistical methods were appropriate. Thematic coding was used to evaluate open-ended questionnaire qualitative data. Students' written reflections were divided into important categories, including how graphic organizers helped them understand expository texts, learning problems, and favorable opinions of graphic organizer use.

3.6. Instrument Validity and Reliability

An educational assessment specialist and EAP instructor verified the summary rubric. Based on comments, performance descriptors were updated to standardize grading and explain evaluation standards. In order to verify inter-rater reliability, two independent raters scored 30 student summaries. Cohen's Kappa showed substantial agreement at 0.76. Two ESP experts examined the student perception questionnaire for content validity and study objectives. A 20-student pilot test assessed clarity and feasibility. Internal consistency was 0.84, indicating strong dependability using Cronbach's Alpha. The questionnaire's psychometric quality was further supported by item-total correlations and item discrimination analyses, which demonstrated that all items contributed to the total score and effectively separated students with varying perceptual levels.

Table 2. Instrument Validity and Reliability Summary

Instrument	Validity Method	Reliability Test	Result	Interpretation
Summarization Rubric	Expert judgment (2 assessors)	Inter-rater reliability (Cohen's Kappa)	0.76	Substantial agreement
Student Questionnaire	Expert review by ESP lecturers	Cronbach's Alpha (pilot, N = 20)	0.84	High internal consistency

4. Results

This chapter presents the findings from an explanatory sequential mixed methods study examining the effectiveness of graphic organizers (GOs) in enhancing students' comprehension of expository texts in an English for Specific Purposes (ESP) context. The study involved 200 undergraduate students from Information Systems and Informatics Engineering programs at Universitas Duta Bangsa Surakarta, Indonesia. Data were collected through structured summarization tasks using Venn Diagrams and T-Charts, assessed via a validated rubric, and supplemented by student perception questionnaires.

4.1. Improving Reading Comprehension Through Text-Structure-Aligned Graphic Organizers

The analysis of summarization scores revealed distinct performance patterns when students engaged with expository texts using graphic organizer scaffolding. Table 3 presents comprehensive descriptive statistics for students' performance across both text structures and overall achievement.

Table 3. Descriptive Statistics of Summary Scores

Statistical Measure	Compare-Contrast Texts (Venn Diagram)	Cause-Effect Texts (T-Chart)	Overall Performance
Mean	3.24	3.67	3.46
Median	3.25	3.75	3.50
Mode	3.00	4.00	3.25
Standard Deviation	0.84	0.71	0.81
Skewness	-0.12	-0.34	-0.18
Kurtosis	-0.45	0.21	-0.28
Minimum Score	1.50	1.75	1.50
Maximum Score	4.75	5.00	5.00
Range	3.25	3.25	3.50

The 5-point summary rubric mean score of 3.46 (SD = 0.81) suggests that students have above-average competency in using graphic organizers (GO) to understand expository texts. With a minor negative skewness (−0.18), the score distribution approximated normalcy, suggesting more students scored above the mean. This suggests that GOs mostly improved students' reading comprehension. T-Charts performed better than Venn Diagrams for cause-and-effect texts ($M = 3.67$, $SD = 0.71$). T-Charts may be more effective than Venn Diagrams for organizing expository content, as cause–effect texts tend to have a lower standard deviation, indicating greater student consistency.

GO helped most students, but reading proficiency and visual tool familiarity varied. Approximately 68% of students scored within one standard deviation of the mean (2.65 to 4.27), demonstrating that GOs helped students of all abilities. To assess how text structure affects graphic organizer-based reading comprehension, a paired samples t-test compared summary results across compare-contrast and cause-effect activities.

Table 4. Paired Samples T-Test Results

Comparison	Mean Difference	SE	t-value	df	p-value	95% CI
Cause-Effect vs Compare-Contrast	0.43	0.062	6.94	199	< 0.001	[0.31, 0.55]

A paired-sample t-test assessed compare-contrast and cause-effect summary scores. Students showed a substantial improvement in summarizing cause-effect texts compared to compare-contrast texts ($t(199) = 6.94$, $p < 0.001$). On a 5-point rubric scale, the mean score difference was 0.43 points, and the 95% confidence range was [0.31, 0.55]. This suggests that ESP students found the T-Chart visual organizer for cause-and-effect readings more natural than the Venn Diagram for compare-contrast interactions. The T-Chart organization's logical and linear flow aligned with students' comprehension schemas for technical processes and system relationships in information technology, presumably contributing to higher cause-effect text performance. However, Venn Diagrams' overlapping nature may raise pupils' cognitive workload as they organize material into several

categories. Cohen's d was calculated to determine the effect size and practical importance of the text type difference.

Table 5. Effect Size Analysis

Comparison	Cohen's d	Effect Size Classification	Interpretation
Cause-Effect vs Compare-Contrast	0.56	Medium	Educationally meaningful improvement

Cohen's d was 0.56 to assess the practical significance of the observed discrepancies. This score shows a medium effect size, showing that these text genres' differences boost pupils' reading comprehension. Cohen's (1988) effect size convention implies that 71% of students surpassed compare-contrast texts when writing cause-and-effect texts with visual organizers. This study has significant educational implications for ESP, as it suggests that the structure of explanatory text and the format of graphic organizers affect learning outcomes. This suggests that educators should employ the T-chart style to teach cause-and-effect relationships in technical literature to EFL students in STEM fields. The medium effect size supports visual organizers as ESP pedagogical aids, especially for students who must learn English while processing complex technical knowledge. This discovery supports the use of structured visual organizers in technical English language programs.

4.2. Patterns of Student Performance in Summary Writing Using Graphic Organizers

The analysis of student performance across the four dimensions of the summarization rubric revealed distinct patterns in reading comprehension skills when supported by graphic organizers. Table 6 presents the distribution of performance levels across accuracy, paraphrasing, focus and relevance, and language structure dimensions.

Table 6. Performance Distribution by Rubric Dimension

Rubric Dimension	Mean Score	SD	Low Performance (1-2)	Moderate Performance (3)	High Performance (4-5)
Accuracy	3.72	0.79	14.0%	28.5%	57.5%
Paraphrasing	2.98	0.93	35.5%	32.0%	32.5%
Focus and Relevance	3.81	0.74	11.5%	31.0%	57.5%
Language Structure	3.34	0.91	24.0%	31.5%	44.5%

Students performed best in the Focus and Relevance dimension ($M = 3.81$, $SD = 0.74$), with 57.5% achieving high scores. This indicates that graphic organizers effectively helped students identify main ideas and maintain topical coherence. Performance in Accuracy was similarly strong ($M = 3.72$, $SD = 0.79$), suggesting that students were able to extract essential content from the texts correctly.

In contrast, Paraphrasing emerged as the weakest area ($M = 2.98$, $SD = 0.93$), with 35.5% of students demonstrating low performance. This finding highlights a significant limitation in students' linguistic competence: while graphic organizers facilitate content organization, they do not automatically develop the skills needed for effective paraphrasing.

To address this challenge, the integration of targeted strategies is recommended. Practical approaches may include:

- Guided paraphrasing exercises using sentence starters or templates to model rephrasing techniques.

- Synonym replacement drills to expand vocabulary and encourage linguistic flexibility.
- Stepwise summarization tasks, where students first extract key ideas and then rewrite them in their own words.
- Peer review activities, allowing students to compare paraphrases and receive constructive feedback.

Finally, performance in the Language Structure dimension was moderate ($M = 3.34$, $SD = 0.91$), with 44.5% achieving high performance. Although graphic organizers supported the organization of ideas, students still faced challenges in producing grammatically coherent summaries. These findings underscore the need for combining visual scaffolding with explicit instruction in both linguistic skills and academic writing strategies.

Figure 1 depicts the comparative performance of three exemplary student summaries across four evaluative dimensions: correctness, focus, paraphrase, and language structure. The exemplary student consistently achieved high scores in accuracy and concentration, demonstrating a thorough comprehension of essential concepts and logical organization, although paraphrasing was somewhat constrained. The student with intermediate performance demonstrated sufficient understanding and structural coherence but relied excessively on the source language, indicating insufficient paraphrasing abilities. Conversely, the underperforming student encountered difficulties in all aspects, especially in identifying essential information, rephrasing material, and ensuring grammatical coherence. The findings suggest that although visual organizers facilitate content organization, they require additional instruction in paraphrasing and academic language to enhance the quality of summaries.

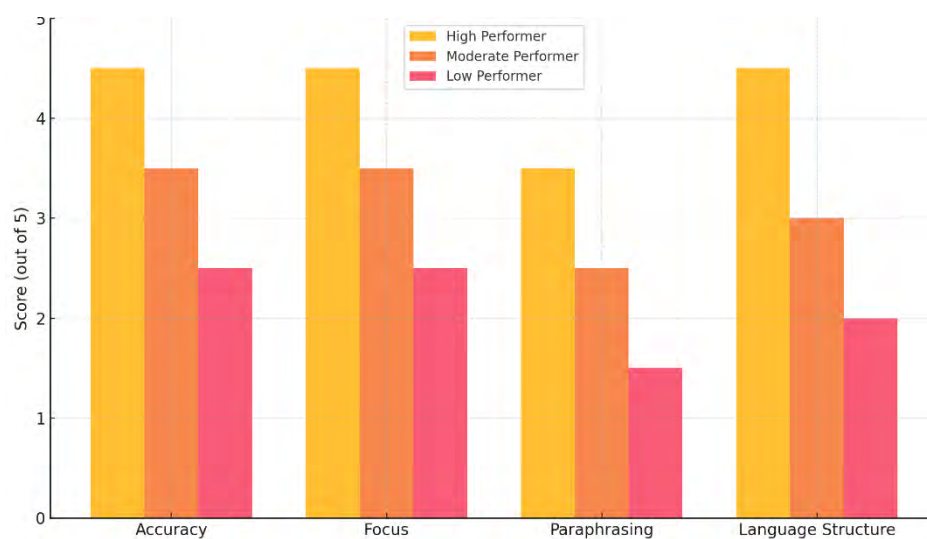


Figure 1. Comparison of Students Performances

4.3. Students' Reflections on Graphic Organizer Use: Benefits, Challenges, and Strategic Awareness

Thematic analysis of 200 open-ended responses revealed three key aspects of students' experiences with graphic organizers (GO) in comprehending expository texts. First, students perceived GO as an practical tool for supporting comprehension through visual structuring, which helped externalize thought processes and reduce cognitive load. T-Charts were especially useful for cause-and-effect texts, while Venn Diagrams posed initial challenges due to overlapping categories, particularly in technical contexts. Second, students reported implementation challenges, including time constraints and uncertainty in categorizing information, especially in compare and contrast texts. However, efficiency improved with practice. Third, students noted enhanced metacognitive

awareness and improved retention, with several recognizing text structures in other contexts and attributing long-term recall to the visual layout of ideas. Representative student responses consistently supported these themes.

Table 7. Representative Student Responses by Theme

Theme	Student Quotes	Frequency Pattern
Cognitive Scaffolding	"The T-chart made me see how network security problems cause specific system failures. Before this, I just read without connecting the relationships."	78% of responses mentioned structural clarity
	"A Venn diagram helped me organize the similarities between programming languages, but it was harder to decide where to put some features."	65% noted categorization benefits with some difficulty
Implementation Challenges	"First time using the T-chart took me almost 30 minutes extra, but by the third test, I could complete it much faster."	82% reported initial time concerns
	"Sometimes I was not sure if a feature belonged in both circles or just one side of the Venn diagram."	71% experienced categorization uncertainty
Metacognitive Enhancement	"Now I automatically look for cause-and-effect patterns when reading technical manuals for my programming class."	69% reported strategy transfer
	"I can remember the cybersecurity concepts better because I can visualize the T-chart in my mind during exams."	74% noted improved retention

Quantitative analysis of perception questionnaire items confirmed the qualitative themes while revealing nuanced differences in student attitudes toward specific aspects of graphic organizer implementation.

Table 8. Student Perception Questionnaire Results

Questionnaire Item	Mean	SD	Strongly Agree/Agree (%)	Neutral (%)	Disagree/Strongly Disagree (%)
GO helped me understand text structure better	4.23	0.71	87.5%	10.0%	2.5%
GO made identifying main ideas easier	4.18	0.74	85.0%	12.5%	2.5%
T-charts were easier to use than Venn diagrams	4.01	0.89	78.5%	15.0%	6.5%
GO instructions were clear and easy to follow	3.94	0.82	76.0%	18.5%	5.5%
GO helped me write better summaries	3.87	0.86	72.5%	21.0%	6.5%
I would use GO in future reading tasks	3.79	0.91	68.0%	24.5%	7.5%
GO reduced my reading comprehension difficulties	3.72	0.94	65.5%	25.5%	9.0%

Thematic analysis of 200 open-ended responses indicated that students perceived graphic organizers as advantageous for enhancing comprehension of expository texts. Numerous individuals

observed that the visual framework elucidated relationships among ideas, diminished cognitive burden, and enhanced concentration on primary concepts. T-Charts were seen as natural for depicting cause-and-effect relationships, especially in technical writings, but Venn Diagrams posed initial difficulties due to overlapping categories. Notwithstanding these challenges, students recognized that graphic organizers enhanced their comprehension of text structure and content arrangement. Several students reported difficulties during early usage, including prolonged reading times and trouble categorizing complex information. Nevertheless, these challenges lessened with practice, indicating enhanced internalization of the strategy. Students showed improved metacognitive awareness and long-term retention, with many seeing structural patterns in various texts and retaining technical material with greater ease. These replies suggest that graphic organizers facilitated both immediate understanding and the enhancement of overarching reading strategies.

5. Discussion

Graphic organizers (GOs) improved students' English for Specific Purposes (ESP) reading comprehension, especially when they fit the text structure. A substantial difference $t(199) = 6.94$, $p < 0.001$ and medium effect size (Cohen's $d = 0.56$) between cause-effect texts using T-Charts ($M = 3.67$) and comparison-contrast texts using Venn Diagrams ($M = 3.24$) revealed better student scores on the former. This shows that GOs' efficacy depends on the format's fit for the information's structure. Cognitive theories, such as Dual Coding Theory, suggest that simultaneous verbal and visual processing can enhance learning, particularly for bilingual learners with limited language proficiency (Wong & Samudra, 2021). This theory may explain this. Visual representations like T-Charts help organize information and improve comprehension by boosting dual processing pathways. Schema Theory posits that current knowledge structures influence the processing and integration of new information (Maxim & Brown, 2023). Text structure-based visualization can help students activate and expand their schemas. The multimedia learning approach emphasizes cognitive load-based instructional design (Ginns & Leppink, 2019; Mayer, 2024) and uses GO as a scaffolding medium to help students navigate complex technical texts. Selecting the right graphic organizer is not only a visual aid; it is a cognitive theory-based instructional method that can improve academic literacy acquisition in technical study programs.

The investigation of rubric dimensions, alongside performance variations attributed to text structure, also uncovered persistent patterns in the quality of students' summaries. The parameters of focus and accuracy yielded the highest results, indicating that the use of graphic organizers (GOs) significantly assisted students in identifying essential concepts and maintaining topic coherence. Nonetheless, the poorest performance was observed in the paraphrasing aspect, since the majority of pupils continued to replicate verbatim from the source material. This indicates that while GOs effectively aid in material comprehension, they do not inherently cultivate sophisticated linguistic abilities such as information reformulation. This conclusion aligns with prior research indicating that paraphrasing skills necessitate explicit teaching. Targeted treatments employing paraphrase guidelines (Yahia & Egbert, 2023) and the notion of grammatical metaphor have demonstrated efficacy in enhancing students' paraphrasing abilities (Walsh Marr, 2019). Moreover, targeted instruction in text structure and reformulation techniques enhances the ability to articulate primary ideas and comprehend reading material (Stevens et al., 2020), while also strengthening academic writing proficiency through knowledge of grammar and vocabulary (Parameswari et al., 2024). Students frequently encounter challenges with paraphrasing owing to constraints in language proficiency, comprehension, and documentation techniques (Flores & Lopez, 2019), hence underscoring the necessity of specialized training. This method enhances academic performance and diminishes cognitive load (Siregar, 2023), and has demonstrated efficacy in legal and linguistic education through the application of explanation and task repetition tactics (Boginskaya, 2020; Khezrlou, 2021). To optimize the efficacy of GO in enhancing comprehensive academic literacy, its

application must be combined with explicit language-focused learning strategies, such as teaching in paraphrasing, academic sentence construction, and contextualized vocabulary enhancement.

A qualitative investigation of open-ended responses and questionnaire data revealed that most students preferred using graphic organizers (GOs) to comprehend expository texts. Students claimed graphic organizers, especially T-Charts, helped them organize knowledge, remember it, and understand complex text structures. This conclusion is supported by elevated mean scores on perception items like “GO facilitated my comprehension of text structure” ($M = 4.23$) and “GO simplified the identification of main ideas” ($M = 4.18$), as well as direct student quotations stating that they could “recall network security concepts due to their ability to visualize the T-Chart during examination. The Dual Coding Theory states that concurrent visual and verbal processing promotes learning efficacy, and these perceptions show how GOs promote cognitive processes and knowledge retention (Wong & Samudra, 2021). Students' experiences identifying text structures outside the classroom demonstrate metacognitive knowledge and transfer of reading methods (Gazali & Baa, 2023). (Balepur et al., 2023) found that GO improves autonomous learning and self-monitoring of comprehension. (Karami, 2020) Also claimed that graphic organizers help EFL readers understand complex texts by visually organizing them. These results demonstrate that GO is more than a reading aid; it fosters deliberate, sustained, and transferable learning processes that can be applied in various academic environments.

The mixed-methods approach employed in this study is a methodological strength, as it integrates qualitative and quantitative data to offer a comprehensive understanding of the efficacy of graphic organizers (GOs). This design is consistent with the sequential exploratory approach that is recommended for the comprehensive evaluation of educational interventions. The emphasis on aligning text structure with the GO format (T-Charts for cause-effect and Venn Diagrams for comparison-contrast) is a practical and theoretical contribution that is still rarely investigated in the ESP literature. This support for cognitive-based instructional design is similar to that of multimedia theory (Mayer, 2024). The rubric-based approach and thematic analysis also enhance the validity of data triangulation, enabling a more comprehensive mapping of student performance and perceptions (Öcal & Önsüz, 2020). Nevertheless, the findings' generalizability is restricted because the participants were exclusively recruited from a single institution and a particular study program. The results on paraphrasing and sentence structure aspects may also be influenced by the limitations of students' English proficiency, which suggests the necessity of incorporating language training into the intervention design (Parameswari et al., 2024; Zein, 2016). Furthermore, the absence of a control group renders it impossible to experimentally isolate the effects of GO, and long-term effects have not been assessed longitudinally (Ginns & Leppink, 2019; Leahy, 2022). These results continue to be pertinent, particularly in terms of offering suggestions for the implementation of visual-based academic literacy teaching practices in higher education contexts of developing countries.

6. Conclusion

This research demonstrates that the reading comprehension of ESP students is substantially enhanced by the use of graphic organizers (GOs) that are aligned with the text structure. T-Charts are more effective than Venn Diagrams in the accurate and focused summation of cause-and-effect texts. GOs also facilitate the identification of primary ideas and relationships between ideas, although paraphrasing skills still require improvement. The significance of incorporating visual approaches in academic literacy learning, particularly in technical study programs, is underscored by student perceptions that GOs support reading strategies and information retention.

7. Suggestion

Universities are encouraged to systematically integrate text-structure-based graphic organizers (GOs) into ESP curricula, accompanied by clear pedagogical guidance for lecturers. Instructors should

anticipate potential difficulties, such as categorizing overlapping information, and provide scaffolding through stepwise demonstrations, guided exercises, and practice sessions. To improve students' paraphrasing skills, structured interventions should be implemented, including guided paraphrasing exercises with sentence starters or templates, synonym and vocabulary drills, stepwise summarization tasks, and peer review sessions for feedback. These strategies not only support the effective use of cognitive scaffolds but also enhance linguistic competence and reading comprehension, aligning with Dual Coding and Schema Theory. Future studies could investigate the long-term effects of such interventions across various disciplines, educational levels, and digital formats, thereby ensuring broader pedagogical applicability.

Declarations

Author Contributions. M.: Literature review, conceptualization. M.M.: methodology, data analysis. S.: review-editing and writing, original manuscript preparation. All authors have read and approved the published version of the article.

Conflicts of Interest. The authors declare that they have no conflict of interest.

Data Availability Statement. Data can be provided upon request.

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